

BlockFlow (IR-ABPS) — Builders Guidebook & Deployment Manual

Smart India Hackathon (SIH 2026)

Problem Statement ID: SIH26027

Problem Statement Title: AI-Powered Automatic Block Planning to Maximize Asset Availability for Train Operations on Indian Railways

Platform: BlockFlow (IR-ABPS)

1. System Requirements & Environment Setup

BlockFlow is engineered to execute completely **air-gapped** on standard Indian Railways divisional workstation hardware without external cloud or GPU requirements.

Minimum Hardware Specifications:

- **Processor:** Intel Core i3 (8th Gen+) or AMD Ryzen 3
- **RAM:** 4 GB DDR4 (BlockFlow consumes <250 MB)
- **Storage:** 500 MB free disk space
- **OS:** Windows 10/11 or Ubuntu Linux 20.04+
- **Network:** Completely offline compatible (Zero external API dependencies)

Software Prerequisites:

- Python 3.11+
 - Modern Web Browser (Google Chrome, Microsoft Edge, or Mozilla Firefox with WebGL enabled)
-

2. Step-by-Step Service Startup Guide

Step 1: Install Python Dependencies

Open PowerShell or Terminal inside the repository root (D:\projects\TTC):

```
pip install -r Backend/requirements.txt
```

Step 2: Start the FastAPI Backend Gateway

```
python -m uvicorn Backend.main:app --host 127.0.0.1 --port 8000
```

- **Interactive API Documentation (Swagger):** <http://127.0.0.1:8000/docs>

- **Health Check Endpoint:** `http://127.0.0.1:8000/api/health`

Step 3: Launch the Control Office Console & 3D Twin

Open your browser and navigate to:

```
http://127.0.0.1:8000
```

This loads `Frontend/block_planner.html` directly via FastAPI's static file mount, rendering:

1. **CesiumJS 3D Railway Digital Twin:** Western Railway 60km suburban corridor (Churchgate to Virar, 29 stations).
2. **Interactive Time-Distance String Graph:** Real-time train trajectories against scheduled joint mega blocks.
3. **Weekly & Monthly Multi-Horizon Schedulers:** Operational crossover diversions and heavy machine rosters.
4. **Chief Controller Telemetry HUD:** Live ACI scores, downtime counters, and 1-click digital grant permits.

Step 4: Run the End-to-End Automated Test Suite

In a separate terminal window:

```
python test_e2e.py
```

Expected output:

```
[PASS] Ingestion Layer: 50 CRIS requisitions validated
[PASS] ACI Engine: 100% Critical P1 flaws scored >= 80
[PASS] HiGHS Simplex Solver: Solved in 5.4ms (0 conflicts)
[PASS] Shadow Bundling: 45% multi-department coordination
[PASS] Crossover Diversion: 0 train cancellations
=====
100% TESTS PASSED - SYSTEM READY FOR DEPLOYMENT
```

3. Universal API Documentation & JSON Payloads

1. Ingest Requisitions & Optimize Blocks

- **Endpoint:** `POST /api/optimize`
- **Sample Request Payload:**

```
{
  "corridor": "Churchgate - Virar",
  "planning_horizon_days": 7,
  "requisitions": [
    {
      "asset_id": "TMS-ENG-1006",
      "department": "CIVIL",
      "km_start": 19.4,
```

```

        "km_end": 21.2,
        "line": "DN_FAST",
        "duration_mins": 180,
        "flaw_type": "USFD_IMR",
        "psr_speed": 30,
        "overdue_days": 4,
        "statutory_form": "IRPWM_284"
    },
    {
        "asset_id": "SMMS-SNT-2012",
        "department": "SNT",
        "km_start": 19.8,
        "km_end": 20.2,
        "line": "DN_FAST",
        "duration_mins": 90,
        "point_no": "104B",
        "statutory_form": "T_351"
    },
    {
        "asset_id": "TDMS-TRD-1003",
        "department": "TRD",
        "km_start": 19.0,
        "km_end": 22.0,
        "line": "DN_FAST",
        "duration_mins": 120,
        "hotspot_temp_c": 84.5,
        "feeding_post": "JOS",
        "statutory_form": "ACTM_203"
    }
]
}

```

- **Sample Response Payload:**

```

{
  "status": "OPTIMAL",
  "solver_latency_ms": 6.8,
  "total_downtime_saved_mins": 195,
  "passenger_cancellations": 0,
  "joint_blocks": [
    {
      "block_id": "JB-2026-04-WR",
      "corridor_section": "Bandra (BA) - Andheri (AND)",
      "line": "DN_FAST",
      "window_start": "01:15",
      "window_end": "04:30",
      "allocated_duration_mins": 195,
      "bundled_departments": ["CIVIL", "SNT", "TRD"],
      "primary_task": "TMS-ENG-1006 (USFD IMR Tamping)",
      "shadow_tasks": ["SMMS-SNT-2012", "TDMS-TRD-1003"],
      "traffic_diversion": {
        "crossover_station": "Bandra (BA)",
        "diverted_trains": ["EMU-90241"],

```

```
    "alternate_line": "DN_SLOW",
    "added_delay_mins": 0
  },
  "statutory_compliance": {
    "earthing_buffer_mins": 15,
    "forms_generated": ["T_351", "T_352", "ACTM_PTW"]
  }
}
]
```

4. Offline Fallback & Live Jury Demo Resilience

To guarantee **zero downtime** during the live SIH jury evaluation round (even if WiFi completely fails):

1. **Local Self-Contained Assets:** All CesiumJS libraries, 3D building models, station coordinates, and satellite imagery tiles are cached locally in `Frontend/`.
2. **Pre-Compiled Simplex Binaries:** SciPy HiGHS runs natively via Python C++ extensions with zero network calls.
3. **Synthetic Live Telemetry Generator:** If live CRIS feeds are disconnected, BlockFlow automatically engages its built-in realistic Western Railway simulation engine, generating live train movements and defect alerts across the 60km corridor.